Physics-Informed Neural Networks for Mechanics Problems

UE24CS352A Machine Learning – Mini-Project, Problem 13 | SRN: PES1UG24CS648

1. Problem Statement

Problem 13: *Data Driven Solutions and Discoveries in Mechanics Using Physics Informed Neural Network* (Zhang, Chen & Yang, Stanford CS229). Many mechanics phenomena obey conservation laws written as differential equations. A physics-informed neural network (PINN) approximates the unknown field with a neural network and embeds the governing equation in the training loss using automatic differentiation. We reproduce the paper's two problem classes: **data-driven solutions** (forward problems: 1D soil consolidation and 2D steady-state heat conduction) and **data-driven discovery** (an inverse problem: identifying the damping coefficient *c* and stiffness *k* of a single-degree-of-freedom oscillator from observations).

2. Dataset Details

No external dataset is used; all data is synthetic and generated in code. Training points are sampled in the problem domain: **collocation (residual) points** where the PDE/ODE is enforced, **boundary/initial points**, and, for the inverse problem only, **observations** computed from the analytical solution with the true parameters c = 0.4, k = 4 (m = u_0 = v_0 = 1). Reference solutions for evaluation: the analytical series solution (consolidation), the analytical damped-oscillation solution (oscillator), and a 101x101 finite-difference solution of the Poisson equation (heat conduction; the paper uses a deal.II finite-element reference, which we replaced with finite differences).

3. Approach

A fully connected tanh network u(x; w), with weights and biases w, approximates the solution. The loss is L = $w_f L_f$ + $w_b L_b$ + $w_d L_d$, the mean-squared PDE residual, boundary/initial-condition mismatch and data mismatch. Derivatives in the residual come from automatic differentiation (PyTorch autograd), so the method is mesh-free. For the inverse problem, *c* and *k* are trainable parameters optimised jointly with the network weights. Because the loss is non-convex, we train with Adam and (for consolidation) refine with L-BFGS. Consolidation is solved in dimensionless form (x, t, p scaled to O(1)), and the output is multiplied by x so that p(0,t) = 0 holds exactly.

4. Implementation Overview (PyTorch)

Script	Network / points	Loss weights, optimiser
pinn_consolidation.py	5x50 tanh, Glorot normal; 500 residual, 250 initial, 125 Neumann points; t in (0, 0.5)	0.25 each; Adam 20,000 epochs (lr 1e-3) then L-BFGS (max 3000 iterations)
pinn_heat.py	4x50 tanh, Glorot uniform; 1500 interior, 500 boundary points on $[-1,1]^2$	0.5 / 0.5; Adam 20,000 epochs (lr 5e-4)
pinn_oscillator.py	3x32 tanh; 50 equispaced observations on [0,15], 100 residual points; c, k initialised to 1	0.25 each (ODE, initial conditions, data); Adam 30,000 epochs (lr 1e-3)

5. Results

Problem	Metric	Result
Oscillator (inverse)	Identified c, k (true 0.4, 4.0)	c = 0.4099 (2.5% off), k = 4.0019 (0.05% off)
Heat conduction (forward)	MSE / max abs. error vs. finite differences	2.54e-6 / 1.33e-2
Consolidation (forward)	MSE / max abs. error vs. analytical, full domain	7.11e-4 / 0.648
	Same, excluding the initial instant (t >= 0.02)	2.35e-4 / 0.140

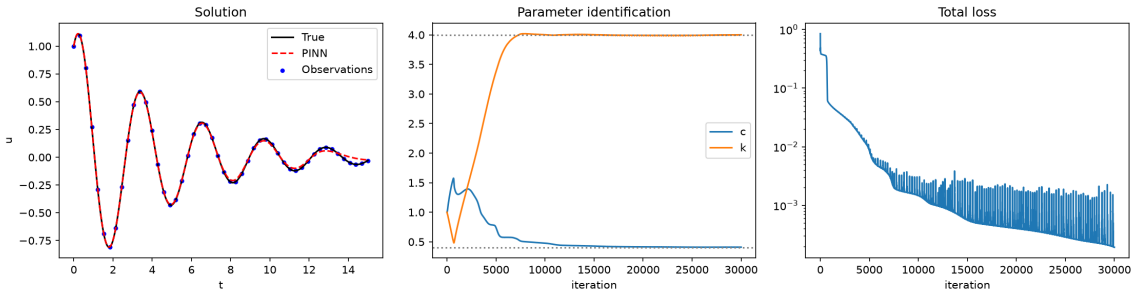


Fig. 1. Oscillator inverse problem: solution vs. observations (left), convergence of c and k to the true values (middle), total loss (right).

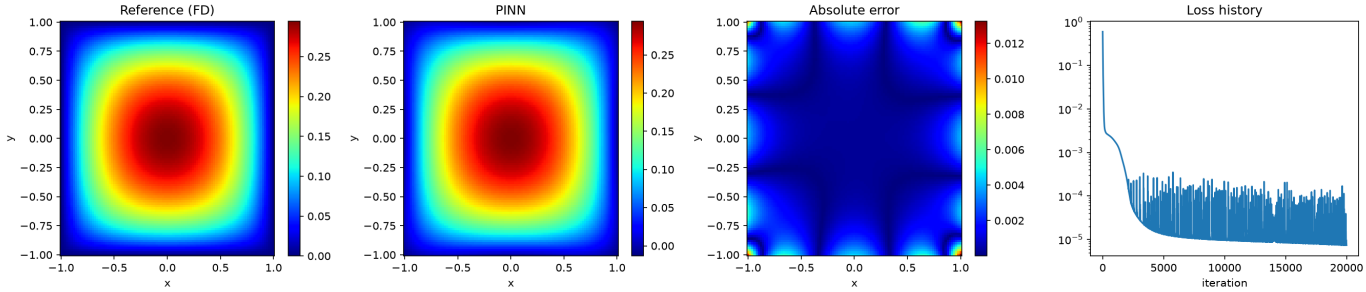


Fig. 2. Heat conduction: finite-difference reference, PINN solution, absolute error (largest near the boundary corners), and loss history.

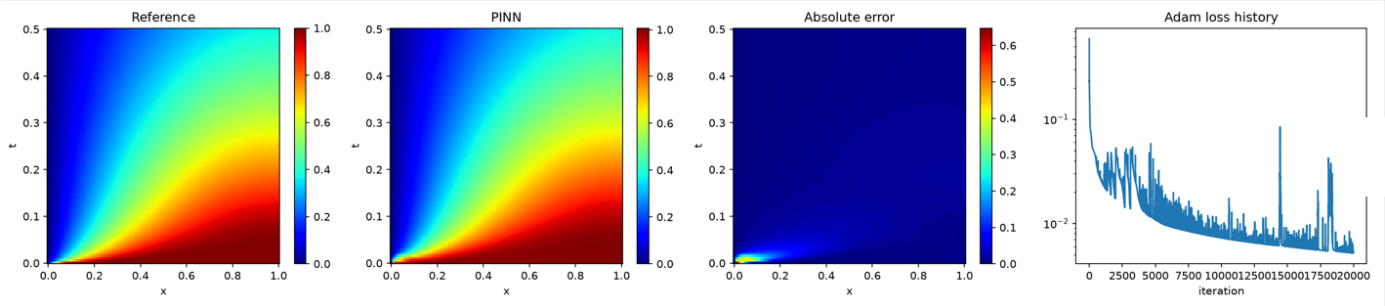


Fig. 3. Consolidation: analytical reference, PINN solution, absolute error (concentrated at x, t near 0), and Adam loss history.

6. Conclusions

The PINN recovered the oscillator parameters to within 2.5% (c) and 0.05% (k) from 50 observations, and reproduced both forward solutions closely without any mesh. For heat conduction the largest error (about 0.013, versus a peak temperature of about 0.29) appears near the boundary because the Dirichlet condition is imposed softly through the loss; a distance-function output transform, as suggested in the paper, would enforce it exactly. For consolidation the error is concentrated at the origin where the solution has a near-discontinuity (initial jump from 1 to 0 at x = 0); away from the initial instant the MSE drops to 2.35e-4. Our errors are somewhat larger than the paper's, most likely because we trained for fewer epochs (20,000-30,000 versus 50,000-100,000), and the oscillator prediction drifts slightly from the true curve at late times (t > 13). Overall, PINNs give a flexible, differentiable surrogate for both solving and calibrating physics models; future work includes adaptive placement of training points near sharp gradients and coupled poromechanics equations.

Code: private GitHub repository (README with setup and run instructions). Reference: Q. Zhang, Y. Chen, Z. Yang, Stanford CS229 project report, 2020.